

Rational Linear Relations and Laplace Equations for Dihedral Modular Graph Functions up to Weight 11

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Abstract

We study rational linear relations among two-loop dihedral modular graph functions $C(a, b, c)$ of weight $w = a + b + c$, together with $E(w)$ and $\zeta(w)$. For the full dihedral basis at weights 3, 5, 7 and 9 the relation space is numerically of dimension exactly one per weight. At weight 9 it is spanned by $960E(9) + \zeta(9) - 2160C(4, 4, 1) - 4320C(4, 3, 2) - 960C(3, 3, 3) = 0$. At weight 11 the relation space was determined only on the sub-basis $C(5, 5, 1), C(5, 4, 2), C(5, 3, 3), C(4, 4, 3), E(11), \zeta(11)$, where it is again one-dimensional. For the full ten-function weight-11 basis the question is undecided. A counting of linearly independent modular graph forms was given in [1], and the weight-9 result may be implied there. We also verify inhomogeneous Laplace equations for $C(3, 1, 1)$, $C(2, 1, 1)$ and $C(2, 2, 1)$. The method is PSLQ on values at random points, followed by an exact leading-Laurent-coefficient check and independent evaluation. All modular-graph-function statements are numerical observations at 4 verifier points, not theorems.

Keywords: modular graph functions, Eisenstein series, string theory one-loop amplitude, Laplace equation, linear relations, low-energy expansion

1 Introduction

Modular graph functions appear as coefficients in the low-energy expansion of the one-loop closed-string amplitude. They satisfy Laplace eigenvalue equations with inhomogeneous terms that are polynomial in non-holomorphic Eisenstein series [2]. Relations of low weight have been proved by an explicit map between graph forms [3], and all dihedral identities at weight seven are in the literature [4]. Systematic basis decompositions of two- and three-point graph forms are available [5]. A counting of linearly independent modular graph forms gives predictions for higher, hitherto unexplored weights [1].

Open question: how many independent rational linear relations exist among all $C(a, b, c)$ of weight w , $E(w)$ and $\zeta(w)$ for weights 3, 5, 7, 9 and 11, and which Laplace equations do the functions satisfy?

Contribution: we determine the relation space numerically on the full dihedral basis at weights 3, 5, 7 and 9. At weight 11 we determine it only on a six-function sub-basis. We verify Laplace equations for three functions, and we reproduce holomorphic identities with Sturm-bound certificates.

Summary of results.

- The weight-by-weight observations show that the relation space has dimension exactly one on the full dihedral basis at weights 3, 5, 7 and 9, with the novelty statuses quoted there.
- A further observation gives the explicit weight-9 relation.
- The weight-11 observation gives the relation and its one-dimensional relation space on the sub-basis only. The full ten-function weight-11 basis is undecided.

- The Laplace observations give equations for $C(3, 1, 1)$, $C(2, 1, 1)$ and $C(2, 2, 1)$.
- The propositions on holomorphic identities certify string-partition-function identities, and a further proposition gives the eta-quotient span table.

2 Model and assumptions

Holomorphic objects. $q = e^{2\pi i\tau}$ is the expansion variable. The Eisenstein series are normalised by $E_k = 1 - (2k/B_k) \sum_n \sigma_{k-1}(n)q^n$, so that $a_0 = 1$. Further building blocks are $\eta(d) = \eta(d\tau)$ (Dedekind eta), $\Delta(d)$ (the discriminant form at $d\tau$), $\theta(d) = \sum_n q^{dn^2}$, the lattice theta series θ_{E8} and θ_{D16} obtained by lattice enumeration, and $j = E_4^3/\Delta$.

Non-holomorphic objects. $\tau = \tau_1 + i\tau_2$ and $p = m\tau + n$. The lattice sums are normalised with weight factor $(\tau_2/(\pi|m\tau + n|^2))^a$:

with weight $w = a + b + c$. $\mathcal{L} = \tau_2^2(\partial_{\tau_1}^2 + \partial_{\tau_2}^2)$ is the hyperbolic Laplacian, and ζ is the Riemann zeta function.

Fixed parameters:

parameter	value
verifier points	4 points, seed 4711, $ \tau_1 \leq 1/2$, $1 \leq \tau_2 \leq 2.5$
working precision	32 digits
tolerance	1e-24 (1e-14 with Laplacian)
finite-difference step	$h = \mathbf{1e-6}$ (fourth-order Laplacian)

3 Method

Relations. Candidate rational relations are obtained by PSLQ on function values at random points. The failed weight-11 attempt on the full basis, at the points examined (6-8 points, up to 64 digits), found no relation. Each candidate is then checked in two ways:

- exact rational arithmetic on the leading Laurent coefficient;
- evaluation at the verifier’s own points, with relative residual below **1e-24**.

Completeness. Completeness of a relation space is judged by a singular-value gap of the normalised value matrix at an independent seed, 4712.

Laplace equations. They are checked at the 4 verifier points at 32 digits, with relative residual at most **1e-14**, using a finite-difference Laplacian, together with an exact leading-coefficient check.

Holomorphic identities. A Sturm-bound certificate is used. For the identity of, the terms lie in $M_2^!(\Gamma_0(4))$ and the coefficients $q^0..q^{11}$ vanish exactly, which exceeds the Sturm bound 1.

Trusted base and status. The trusted base is exact rational arithmetic for leading coefficients and Sturm certificates, plus the verifier’s 32-digit evaluation of modular graph functions. Everything about modular graph functions beyond the exact leading coefficient is a numerical observation at the 4 verifier points, never a theorem. The weight-11 sub-basis came from a human-guided exploration and was then verified by the code verifier. Everything is re-run with the command in the Declarations.

The work was carried out by an automated, verifier-gated laboratory in which every statement reported here was accepted only after an independent code check; its organisation is described in Appendix A.

4 Results

4.1 Relation spaces (numerical observations)

Numerical observation 1 (Weight 3). Numerically, the rational linear relations among $C(1, 1, 1), E(3), \zeta(3)$ form a space of dimension exactly 1, spanned by $C(1, 1, 1) - E(3) - \zeta(3) = 0$.

Certificate. The relation holds to $1\text{e-}24$ at verifier points with exact leading Laurent coefficient. Completeness follows from the singular-value gap in Table 1. Novelty status: inconclusive; equivalent searches disagree (2 equivalent checks, 1 with a literature hit, 1 without), the closest statement being arXiv:1603.00839, and the result may be known or implied there [3].

Numerical observation 2 (Weight 5). Numerically, the relation space among $C(3, 1, 1), C(2, 2, 1), E(5), \zeta(5)$ has dimension exactly 1, spanned by $30C(2, 2, 1) - 12E(5) - \zeta(5) = 0$.

Certificate. The relation holds to $1\text{e-}24$ with exact leading coefficient, and the gap is given in Table 1. Novelty status: inconclusive; equivalent searches disagree (2 equivalent checks, 1 with a literature hit, 1 without), closest statement arXiv:1603.00839v4, and the result may be known or implied there [6].

Numerical observation 3 (Weight 7). Numerically, the relation space among $C(5, 1, 1), C(4, 2, 1), C(3, 3, 1), C(3, 3, 3)$ has dimension exactly 1, spanned by $252C(3, 3, 1) + 252C(3, 2, 2) - 108E(7) - \zeta(7) = 0$.

Certificate. The relation holds to $1\text{e-}24$ with exact leading coefficient, and the gap is given in Table 1. Novelty status: already stated in the literature (arXiv:1608.04393v3) [4].

Numerical observation 4 (Weight 9, relation space). Numerically, the relation space among $C(7, 1, 1), C(6, 2, 1), C(5, 3, 1), C(5, 2, 2), C(4, 4, 1), C(4, 3, 2), C(3, 3, 3), E(9), \zeta(9)$ has dimension exactly 1, spanned by $-2160C(4, 4, 1) - 4320C(4, 3, 2) - 960C(3, 3, 3) + 960E(9) + \zeta(9) = 0$.

Certificate. The relation holds to $1\text{e-}24$ with exact leading coefficient. The gap is given in Table 1. Novelty status: inconclusive; equivalent searches disagree (3 equivalent checks, 1 with a literature hit, 2 without). The closest statement is arXiv:2004.05156v2, which counts linearly independent modular graph forms, and the result may be known or implied there [1].

Numerical observation 5 (Weight 9, explicit relation). Numerically (32 digits, 4 verifier-chosen points, relative residual at most $1\text{e-}24$), $960E(9) + \zeta(9) - 2160C(4, 4, 1) - 4320C(4, 3, 2) - 960C(3, 3, 3) = 0$ as functions of τ .

Certificate. The exact leading coefficient of y^9 vanishes, and the maximal relative residual at the verifier points is $8.33\text{e-}44$. Novelty status: inconclusive; equivalent searches disagree (3 equivalent checks, 1 with a literature hit, 2 without), closest statement arXiv:2004.05156v2, and the result may be known or implied there [1].

Numerical observation 6 (Weight 11, sub-basis only). Numerically, the relation space among $C(5, 5, 1), C(5, 4, 2), C(5, 3, 3), C(4, 4, 3), E(11), \zeta(11)$ has dimension exactly 1, spanned by $19008C(5, 5, 1) + 38016C(5, 4, 2) + 19008C(5, 3, 3) + 28512C(4, 4, 3) - 8640E(11) - \zeta(11) = 0$. This concerns the sub-basis only. For the full ten-function basis (ten $C(a, b, c), E(11), \zeta(11)$) the question is undecided: the PSLQ attempt, with up to 64 digits at the points examined, found nothing and the run was aborted.

Certificate. The relation holds at 4 verifier points at 32 digits with relative residual at most $1\text{e-}24$ (maximum $1.21\text{e-}43$), and the exact leading coefficient of y^{11} vanishes. Completeness on the sub-basis follows from the gap in Table 1. Novelty status: not found in a targeted search of 85 abstracts on 2026-10-04 for the relation, and not found in a targeted search of 121 abstracts on 2026-10-04 for the relation space.

Table 1. Singular values of the normalised value matrices (seed 4712); one singular value is below $1\text{e-}24$ in every row.

weight	basis	matrix	singular values $> 1\text{e-}8$	smallest of these
3	full	6x3	2	0.731
5	full	7x4	3	0.0372
7	full	9x6	5	0.000127
9	full	12x9	8	8.6e-7
11	sub-basis	9x6	5	4.41e-6

4.2 Laplace equations (numerical observations)

Table 2. Laplace equations at 32 digits, 4 verifier-chosen points, relative residual bound $1\text{e-}14$.

equation (as functions of τ)	max. residual
$\mathcal{L}[C(3, 1, 1)] - 6C(3, 1, 1) - \frac{86}{5}E(5) + 4E(2)E(3) - \frac{1}{10}\zeta(5) = 0$	1.63e-25
$\mathcal{L}[C(2, 1, 1)] - 2C(2, 1, 1) - 9E(4) + E(2)E(2) = 0$	2.39e-25
$\mathcal{L}[C(2, 2, 1)] - 8E(5) = 0$	6.99e-26
$\mathcal{L}[C(2, 2, 1)] - 20C(2, 2, 1) + \frac{2}{3}\zeta(5) = 0$	6.74e-26

Numerical observation 7 (Laplace equation for $C(3, 1, 1)$). $\mathcal{L}[C(3, 1, 1)] - 6C(3, 1, 1) - \frac{86}{5}E(5) + 4E(2)E(3) - \frac{1}{10}\zeta(5) = 0$.

Certificate. The exact leading coefficient of y^5 vanishes, with numerical residual as in Table 2. Novelty status: already stated in the literature (arXiv:1608.04393v3) [4].

Numerical observation 8 (Laplace equation for $C(2, 1, 1)$). $\mathcal{L}[C(2, 1, 1)] - 2C(2, 1, 1) - 9E(4) + E(2)E(2) = 0$.

Certificate. The exact leading coefficient of y^4 vanishes, with numerical residual as in Table 2. Novelty status: already stated in the literature (arXiv:1502.06698) [2].

Numerical observation 9 (Source-only equation for $C(2, 2, 1)$). $\mathcal{L}[C(2, 2, 1)] - 8E(5) = 0$.

Certificate. The exact leading coefficient of y^5 vanishes, with numerical residual as in Table 2. Novelty status: already stated in the literature (doi:10.1088/0264-9381/33/23/235011) [7].

Numerical observation 10 (Eigenvalue equation for $C(2, 2, 1)$). $\mathcal{L}[C(2, 2, 1)] - 20C(2, 2, 1) + \frac{2}{3}\zeta(5) = 0$.

Certificate. The exact leading coefficient of y^5 vanishes, with numerical residual as in Table 2. Novelty status: already stated in the literature (arXiv:1502.06698) [2].

4.3 Supporting statement and examples

Numerical observation 11 (Weight 7, pointwise). Numerically (32 digits, 4 verifier points, relative residual at most $1\text{e-}24$), $252C(3, 3, 1) + 252C(3, 2, 2) - 108E(7) - \zeta(7) = 0$ as functions of τ .

Certificate. The exact leading coefficient of y^7 vanishes, and the maximal residual is $1.59\text{e-}43$. Status: already stated in the literature (arXiv:2007.05476v2) [5].

Example 12 (Weight 3). Numerically, $C(1, 1, 1) - E(3) - \zeta(3) = 0$ as functions of τ , with maximal relative residual 5.13e-44 and vanishing exact leading coefficient of y^3 .

Example 13 (Weight 5). Numerically, $30C(2, 2, 1) - 12E(5) - \zeta(5) = 0$ as functions of τ , with maximal relative residual 1.57e-43 and vanishing exact leading coefficient of y^5 .

4.4 Holomorphic identities from string partition functions

These identities are the holomorphic building blocks of string partition functions: the Jacobi identity as an eta-quotient identity, the two heterotic lattices, and the discriminant. Each is certified exactly by a Sturm bound. They are certified statements, not numerical observations.

Proposition 14 (Jacobi identity on $\Gamma_0(4)$). (i) $\theta(1)^4 - \eta(2)^{20}\eta(1)^{-8}\eta(4)^{-8} = 0$ (weight 2).
(ii) $\theta(1)^2 - \eta(2)^{10}\eta(1)^{-4}\eta(4)^{-4} = 0$ (weight 1).

Proposition 15 (Heterotic lattices on $\Gamma_0(1)$). $\theta_{E8} - E_4 = 0$; $\theta_{E8}^2 - \theta_{D16} = 0$; $\theta_{E8}^2 - E_4^2 = 0$; $\theta_{E8}^2 - E_8 = 0$.

Proposition 16 (Discriminant and j on $\Gamma_0(1)$). $1728\Delta(1) - E_4^3 + E_6^2 = 0$, and $j\Delta(1) - E_4^3 = 0$.

Certificate. In each case all terms lie in $M_k^!(\Gamma_0(N))$ of equal weight, and the coefficients vanish exactly up to the stated order, which exceeds the Sturm bound (Table 3).

Table 3. Sturm certificates.

identity	weight	index	vanishing q -coefficients	Sturm bound
$\theta(1)^4$ vs. eta quotient	2	6	$q^0 \dots q^{11}$	1
$\theta(1)^2$ vs. eta quotient	1	6	$q^0 \dots q^{10}$	1/2
$\theta_{E8} = E_4$	4	1	$q^0 \dots q^{10}$	1/3
$\theta_{E8}^2 = \theta_{D16}, E_4^2,$ E_8	8	1	$q^0 \dots q^{10}$	2/3
$1728\Delta = E_4^3 - E_6^2$	12	1	$q^0 \dots q^{11}$	1
$j\Delta = E_4^3$	12	1	$q^0 \dots q^{11}$	1

Novelty status: the Jacobi identities are already stated in the literature (arXiv:2607.26471v1) [8]. The discriminant formula is already stated in the literature (arXiv:1505.06042v1) [9]. The lattice identities and $j\Delta = E_4^3$ are reproductions of classical results, preregistered as validation anchors without any claim of novelty.

Proposition 17 (Eta-quotient span table). *For all levels $N \leq 30$, holomorphic eta quotients with trivial character span $M_4(\Gamma_0(N))$ exactly for $N \in [2, 4, 5, 6, 8, 9, 10, 12, 14, 15, 16, 18, 20, 22, 24, 25, 26, 27, 28, 30]$ and not for $N \in [1, 3, 7, 11, 13, 17, 19, 21, 23, 29]$ (exact, exhaustive).*

Certificate. The span of eta quotients was computed exactly for each level and compared with the space dimension. Novelty status: not found in a targeted search of 33 abstracts on 2026-10-04.

5 Negative results

- **Weight 6 (undecided).** The question concerned relations among $C(4, 1, 1), C(3, 2, 1), C(2, 2, 2), E(6), E(2)$. No claim passed the verifier. The value matrix (12x9) has no singular value below $1\text{e-}24$ and a smallest large singular value of $2.44\text{e-}8$. The singular-value gap criterion is therefore not met, and the dimension remains undecided.

- **Weight 11, first attempt with the full basis.** PSLQ over the full 12-element basis (ten $C(a, b, c)$, $E(11)$, $\zeta(11)$), at the points examined (6-8 points, up to 64 digits), found no relation. The run was aborted. The suspected cause is precision that is too low relative to the coefficient height. This is why the weight-11 result is restricted to the sub-basis.
- **First eta-span run.** No claim passed the verifier. The failure came from a verifier capacity limit, which was fixed, and the table above was then obtained.

6 Discussion, limitations and open questions

The relation space has dimension exactly one at weights 3, 5, 7 and 9 on the full dihedral basis. Each relation has the form (combination of C) = (rational) $E(w)$ + (rational) $\zeta(w)$. The weight-11 relation has the same shape on a six-function sub-basis. All statements on modular graph functions are numerical observations at 4 verifier points, with an exact check only of the leading Laurent coefficient.

Open questions:

- (i) What is the relation-space dimension for the full ten-function weight-11 basis? Higher precision or a different method is needed.
- (ii) What is the dimension at weight 6, where the gap criterion was not met?
- (iii) Is the weight-9 result implied by the counting of [1] (status: inconclusive)?
- (iv) Can the numerical relations be turned into proofs, as done for low weight in [3] [4]?

7 Declarations

AI usage. The results were produced and checked by an automated laboratory; every statement was verified by code.

Code and data availability. Repository <https://github.com/NinjaTurtlesHackathons/Hacknation>, branch `claudemodulformen-paper`, directory `projects/modular`. Reproduce all certificates with `python -m asd.recheck modular` and rebuild the paper with `python -m asd.paper --domain modular`.

Competing interests. None.

A Agentic laboratory

The laboratory ran 8 rounds with 24 verified statements and 3 negative results. Every round was preregistered before the experiment (`prereg.md`). Several agents with distinct roles proposed candidates (for example numerical, skeptical and theoretical roles, as listed in the failed attempts). Only statements accepted by the code verifier were kept. Counter-checks (adversarial tests) numbered 30 in total: 2 passed, 27 did not pass, and 1 was not executable. “Did not pass” means the false variant was rejected, which is the intended outcome. The two that passed did not contradict the statements.

B Provenance

The provenance table mapping every statement to its evidence follows; it is generated automatically.

Statement	Evidence (claim ids)	Level
Section: Introduction	C-modular-R1, C-modular-R1b, C-modular-R1c, C-modular-R1d, C-modular-R4a, C-modular-R4b, C-modular-R4c, C-modular-R4d, C-modular-R5a, C-modular-R5g, C-modular-R6a, C-modular-R7b, C-modular-R8, C-neg3	computed_rigorous, observed
Section: Model and assumptions	C-modell	computed_rigorous
Section: Method	C-modell, C-modular-R1, C-modular-R4a, C-modular-R5a, C-modular-R7b, C-neg3, C-verfuegbarkeit	computed_rigorous, observed
Observation 1 (Weight 3)	C-modular-R1	observed
Section: Relation spaces (numerical observations)	C-modular-R1, C-modular-R1b, C-modular-R1c, C-modular-R1d, C-modular-R6a, C-modular-R6e, C-modular-R7a, C-modular-R7b	observed
Observation 2 (Weight 5)	C-modular-R1b	observed
Observation 3 (Weight 7)	C-modular-R1c	observed
Observation 4 (Weight 9, relation space)	C-modular-R1d, C-modular-R6e	observed
Observation 5 (Weight 9, explicit relation)	C-modular-R6a	observed
Observation 6 (Weight 11, sub-basis only)	C-modular-R7b, C-neg3	observed
Section: Laplace equations (numerical observations)	C-modular-R4a, C-modular-R4b, C-modular-R4c, C-modular-R4d	observed
Observation 7 (Laplace equation for $C(3, 1, 1)$)	C-modular-R4a	observed
Observation 8 (Laplace equation for $C(2, 1, 1)$)	C-modular-R4b	observed
Observation 9 (Source-only equation for $C(2, 2, 1)$)	C-modular-R4c	observed
Observation 10 (Eigenvalue equation for $C(2, 2, 1)$)	C-modular-R4d	observed

Statement	Evidence (claim ids)	Level
Observation 11 (Weight 7, pointwise)	C-modular-R6d	observed
Section: Supporting statement and examples	C-modular-R6d	observed
Example 1 (Weight 3)	C-modular-R6b	observed
Example 2 (Weight 5)	C-modular-R6c	observed
Section: Holomorphic identities from string partition functions	C-modular-R5a, C-modular-R5b, C-modular-R5c, C-modular-R5d, C-modular-R5e, C-modular-R5f, C-modular-R5g, C-modular-R5h, C-modular-R8	computed_rigorous
Proposition 1 (Jacobi identity on $\Gamma_0(4)$)	C-modular-R5a, C-modular-R5b	computed_rigorous
Proposition 2 (Heterotic lattices on $\Gamma_0(1)$)	C-modular-R5c, C-modular-R5d, C-modular-R5e, C-modular-R5f	computed_rigorous
Proposition 3 (Discriminant and j on $\Gamma_0(1)$)	C-modular-R5g, C-modular-R5h	computed_rigorous
Proposition 4 (Eta-quotient span table)	C-modular-R8	computed_rigorous
Section: Negative results	C-modular-R7b, C-modular-R8, C-neg1, C-neg2, C-neg3	computed_rigorous, observed
Section: Discussion, limitations and open questions	C-modell, C-modular-R1, C-modular-R1b, C-modular-R1c, C-modular-R1d, C-modular-R7b, C-neg1, C-neg3	computed_rigorous, observed
Section: Declarations	C-verfuegbarkeit	observed
Section: Appendix A: Agentic laboratory	C-methode, C-modular-R1c-RT1, C-modular-R5g-RT1, C-neg1, C-redteam	computed_rigorous, observed

Statement	Evidence (claim ids)	Level
Section: Appendix C: Verifier self-test	C-modular-R4a, C-modular-R4b, C-modular-R4c, C-modular-R4d, C-modular-R4d-RT1, C-modular-R5a-RT1, C-modular-R5b-RT1, C-modular-R5b-RT2, C-modular-R6a, C-modular-R6b, C-modular-R6c, C-modular-R6d, C-modular-R6d-RT2, C-modular-R7a, C-modular-R7a-RT2, C-modular-R8, C-redteam	computed_rigorous, observed

The complete automatic checking protocol of the writing process is stored as `pruefprotokoll.json` in the project directory of the repository.

C Verifier self-test

The verifier is tested with deliberately false variants, which it must reject. Examples:

- factor 1727 instead of 1728 in the discriminant identity, and 1729 instead of 1728;
- q^1 -coefficient 481 instead of 480 for θ_{D16} ;
- $\zeta(5)$ coefficient $1/5$ instead of $1/10$, which gives residual $1.04\text{e-}01$;
- $\zeta(9)$ coefficient 2 instead of 1;
- a sign flip of the $\zeta(11)$ term, which gives residual $5.71\text{e-}01$.

All of these were rejected, so the verifier distinguishes exact leading coefficients and tight residuals from near-misses.

References

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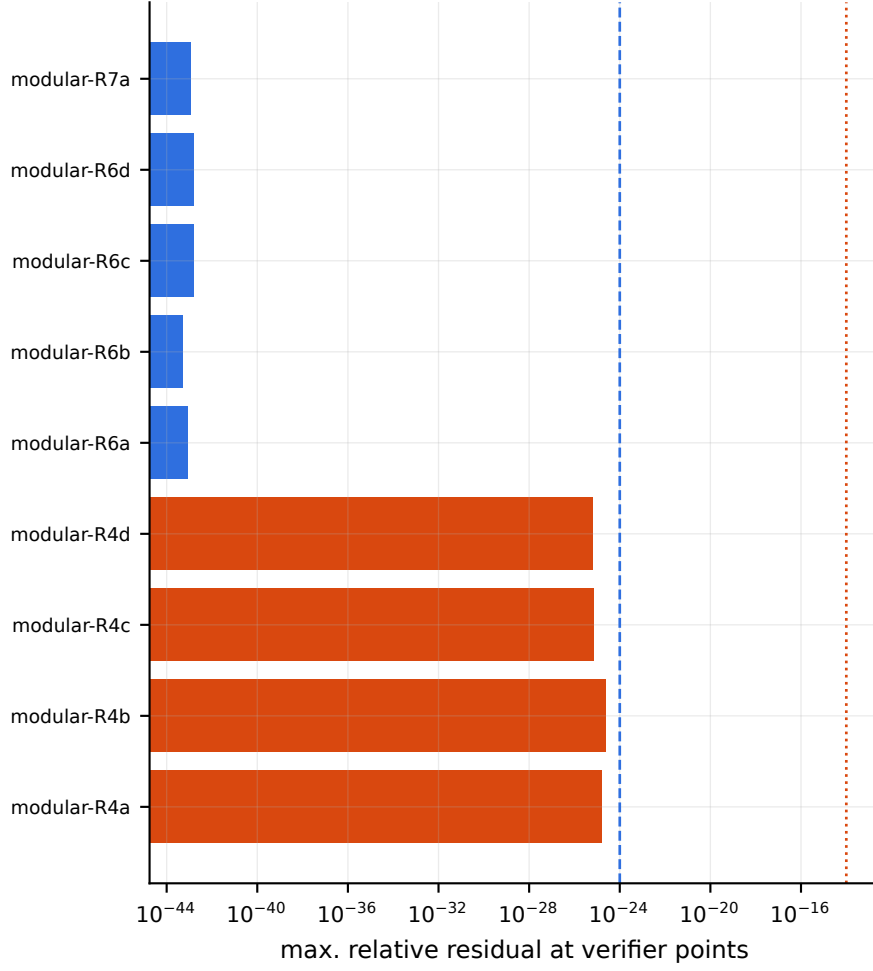


Figure 1: Numerical verification margins: largest relative residual of each confirmed relation among modular graph functions at the verifier’s own random points (32 digits). Dashed: acceptance threshold 10^{-24} ; dotted: threshold 10^{-14} for relations containing a Laplacian (orange).

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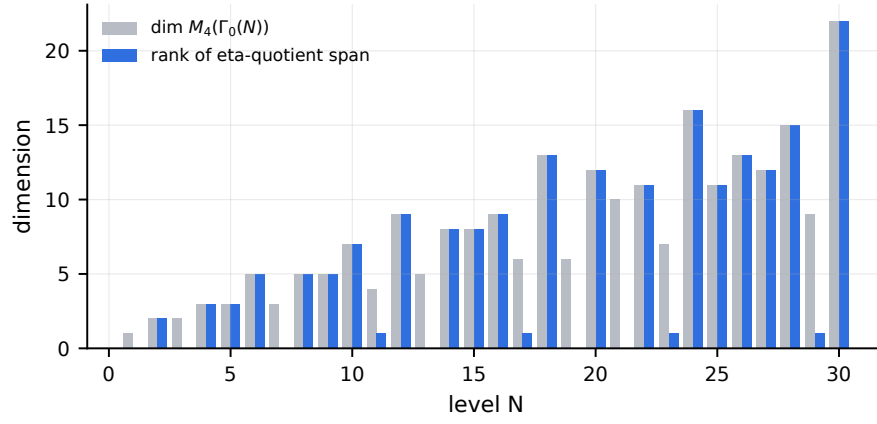


Figure 2: Exact comparison, for every level $N \leq 30$, of $\dim M_4(\Gamma_0(N))$ (grey) with the rank of the span of all holomorphic eta quotients of weight 4 with trivial character (blue).